



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment-1.3

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Semester: 5th

Date of Performance: 19/08/2025

Subject Name: DAA

Subject Code: 23CSH-301

1. **Aim:** Code to find frequency of elements in a given array in $O(n)$ time complexity.
2. **Objective:** To find frequency of elements in a given array in $O(n)$ time complexity.
3. **Input/Apparatus Used:** In this program, HashMap concept is used in order to get less complexity.

4. Procedure/Algorithm: Pseudocode:

Algorithm to find the frequency of each element of an array :

- Input the number of elements of an array.
- Input the array elements.
- Create another array to store the frequency of elements.
- Traverse the input array and update the count of the elements in the frequency array.
- Print the frequency array which displays the frequency of all the elements of the array.

Algorithm to find the frequency of each element of an array using hashing

- Input the number of elements of the array.
- Input the array elements.
- Create a hash table and update the element in one column and its frequency in the other column.
- Print the element along with its frequency.



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Algorithm :

1. Create an extra space of size n (hm), use it as a HashMap.
2. Traverse the array from start to end.
3. For every element update $hm[array[i]-1]$, i.e. $hm[array[i]-1]++$
4. Run a loop from 0 to n and print $hm[array[i]-1]$ along with the index i

Sample Input:

- 1). $arr[] = \{10, 20, 20, 10, 10, 20, 5, 20\}$
- 2). $arr[] = \{10, 20, 20\}$

Sample Output:

- 1).

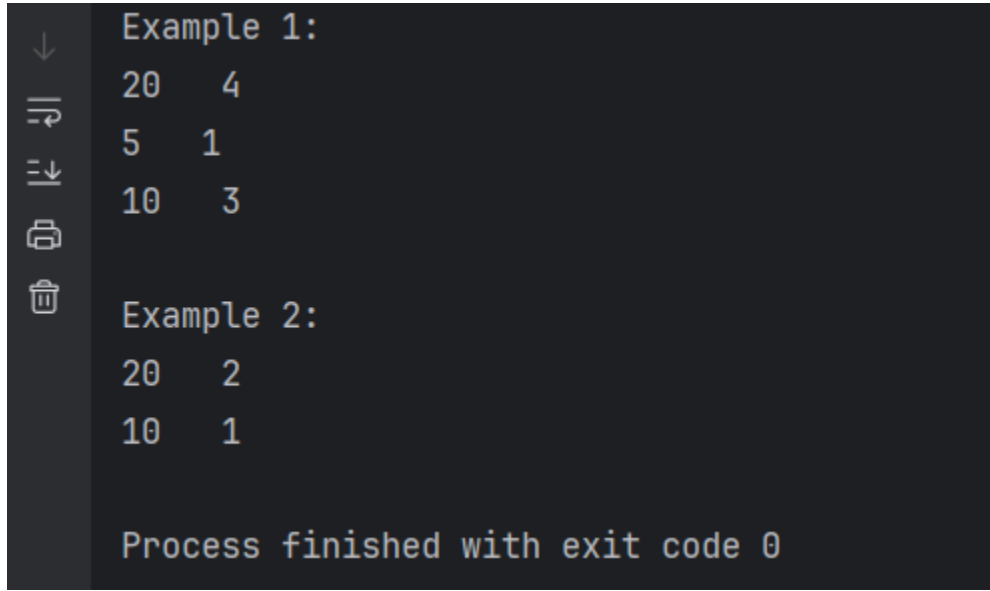
10	3
20	4
5	1
- 2).

10	1
20	2

5. Code:

```
3 import java.util.HashMap;
4 import java.util.Map;
5
6 class FrequencyCounter {
7
8     @static void countFrequency(int arr[]) { 2 usages
9         HashMap<Integer, Integer> freqMap = new HashMap<>();
10
11         for (int num : arr) {
12             freqMap.put(num, freqMap.getOrDefault(num, defaultValue: 0) + 1);
13         }
14
15         for (Map.Entry<Integer, Integer> entry : freqMap.entrySet()) {
16             System.out.println(entry.getKey() + " " + entry.getValue());
17         }
18     }
19
20
21     public static void main(String[] args) {
22         int arr1[] = {10, 20, 20, 10, 10, 20, 5, 20};
23         int arr2[] = {10, 20, 20};
24
25         System.out.println("Example 1:");
26         countFrequency(arr1);
27
28         System.out.println("\nExample 2:");
29         countFrequency(arr2);
30     }
31 }
```

6. Output:



```
Example 1:
20    4
5     1
10    3

Example 2:
20    2
10    1

Process finished with exit code 0
```